

Generative Scenario: Domain Randomization for Robot Vision

Group 12

Abstract—Robotic vision models often fail under distribution shifts because collecting and labeling real-world videos that cover diverse environments, objects, and interaction dynamics is expensive and risky. We present a practical pipeline for *generative semantic augmentation* that transforms an input video into new, logically consistent training samples by combining temporally-aware object segmentation with mask-constrained video generation. Our system integrates automatic prompt proposal, stable spatiotemporal segmentation, and diffusion-transformer-based latent-space video inpainting that preserves motion and background consistency. To enrich and validate the generated data, we further introduce a focused VideoQA module that produces structured textual descriptions of object interactions. Our results demonstrate improved foreground-background disentanglement compared to conventional inpainting-based pipelines, enabling physically plausible object replacement without temporal ghosting or background corruption.

Index Terms—robot vision, domain randomization, video generation, segmentation, diffusion transformer, semantic augmentation, object swapping, video question answering

I. INTRODUCTION

Robotic vision systems frequently fail under distribution shifts because collecting and labeling real-world videos that cover diverse environments, objects, and interaction dynamics is expensive, time-consuming, and sometimes unsafe [1]. As a result, models trained on limited laboratory data often struggle to generalize to complex real-world scenarios [2].

Traditional data augmentation increases low-level appearance diversity but rarely alters semantic content [3]. In contrast, video-based robot perception requires meaningful variation in *what* happens in a scene (object identity, attributes, and interaction context), while maintaining temporal and physical consistency. Naively editing videos often breaks motion continuity or contaminates static backgrounds, leading to unrealistic training data [4].

Our goal is to create a scalable pipeline that uniquely generates semantically diverse yet temporally and physically consistent training videos by combining generative object replacement with motion and background preservation, bridging the domain gap without exhaustive real-world data collection.

Contributions.

- We develop a novel generative semantic augmentation pipeline for video-based robotic vision that uniquely detects, segments, and generatively swaps objects while preserving motion and background stability.
- We identify subject–environment entanglement as a critical failure mode in conventional video inpainting and introduce a refined two-stage strategy that decouples seg-

mentation from generation, a step not found in standard pipelines.

- We introduce a focused VideoQA module that produces fine-grained interaction descriptions, enabling richer metadata and a potential validation loop, which differentiates our approach from existing techniques.

II. RELATED WORK

A. Domain Randomization and Video Augmentation

Domain randomization improves generalization by exposing models to diverse visual appearances and environmental conditions during training, such as variations in object styles, textures, colors, and backgrounds [5]. Prior work has explored synthetic data generation, sim-to-real transfer, and appearance-level randomization to mitigate the domain gap between training and deployment environments [6].

While effective for static perception tasks, these approaches are often insufficient for interaction-heavy video understanding. Naive perturbations that alter appearance without respecting temporal structure can disrupt motion continuity and violate semantic plausibility. As a result, models trained on such data may struggle to learn coherent action dynamics, highlighting the need for augmentation strategies that preserve both temporal coherence and semantic correctness in video sequences [7].

B. Temporally-Aware Video Segmentation

The Segment Anything Model (SAM) demonstrates strong zero-shot segmentation performance on static images but lacks temporal awareness when applied directly to video [8]. SAM-I2V [9] addresses this limitation by extending SAM into the temporal domain while retaining its frozen image encoder, thereby preserving robust 2D feature representations. Temporal coherence is introduced through lightweight adapter modules, resulting in an overall increase in training cost of approximately 0.2% compared to training fully video-native segmentation models. This design makes SAM-I2V particularly suitable for video pipelines that require temporally stable masks without prohibitive computational overhead. SAM-I2V achieves spatiotemporal consistency through three core components. The *Temporal Feature Integrator* fuses high-resolution spatial features from the current frame with motion-aware features from neighboring frames, producing temporally enriched embeddings. The *Memory Selective Associator* mitigates memory explosion in long sequences by retaining only salient long-term appearance cues and short-term motion

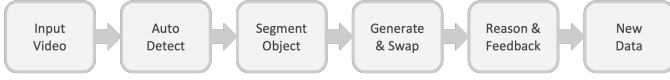


Fig. 1: Pipeline overview: input video → detect & segment → generate & swap → new video with VideoQA description.

trajectories, which are integrated via Transformer-based cross-attention. Finally, the *Memory Prompt Generator* converts stored temporal information into dense prompts that guide the mask decoder, ensuring stable and coherent segmentation across frames.

C. Diffusion-Based Video Generation

Diffusion models have been extended from images to videos by incorporating temporal modules and conditioning mechanisms [10]. However, many video inpainting pipelines still suffer from background warping and temporal artifacts due to foreground–background entanglement during the denoising process [11]. These issues are particularly problematic for object-centric video editing, where preserving static background structure is essential.

Transformer-based diffusion backbones with spatiotemporal latent representations offer improved scalability and temporal modeling capacity [12]. Wan 2.1 [13] represents a paradigm shift from U-Net-based architectures toward a scalable Diffusion Transformer (DiT) design optimized for high-dimensional video synthesis. A key component of Wan 2.1 is its 3D causal Variational Autoencoder, which encodes video into a compact spatiotemporal latent space while strictly enforcing causal ordering. By preventing information leakage from future frames to past frames, this design preserves motion consistency and reduces VRAM usage by approximately 60%, enabling efficient generation of high-fidelity video sequences.

D. Vision-Language Models and Video Question Answering

Vision-language models enable joint reasoning over visual and textual modalities and have shown strong performance in image and video understanding tasks [14]. In the video domain, VideoQA systems generate structured descriptions of actions and interactions, providing semantic supervision beyond raw visual signals.

Recent large-scale models such as Qwen-VL [15] and its video variants demonstrate strong capabilities in temporal reasoning and fine-grained action description. In this work, we leverage a Qwen-based VideoQA model to analyze object-centric video clips and generate detailed textual descriptions of interactions. These descriptions serve as cognitive feedback for annotation, validation, and potentially filtering implausible generations, complementing visual augmentation with structured semantic information.

III. METHODS

A. Core Pipeline

We introduce **Generative Semantic Augmentation** (see Fig. 1). This domain randomization strategy modifies the

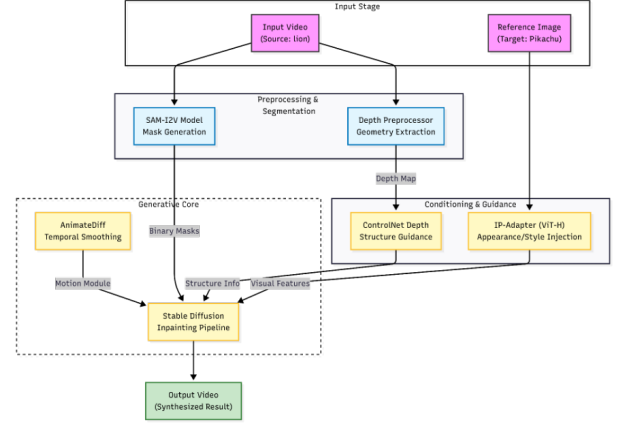


Fig. 2: Object swapping - First Structure

appearance or identity of foreground objects while preserving the behavioral logic of the original video.

- 1) **Input Video:** The process begins with a source video containing an object or action of interest.
- 2) **Auto Detect:** A model automatically detects objects in the video, providing bounding boxes as initial prompts.
- 3) **Segment Object:** The bounding boxes are passed to SAM-I2V, which generates precise and temporally consistent segmentation masks for the target object.
- 4) **Generate & Swap:** The masked region is replaced with a new synthetically generated object, guided by text prompts.
- 5) **Reason & Feedback:** The generated video is analyzed by a VideoQA pipeline to produce a detailed textual description of the interaction.
- 6) **New Data:** The output consists of a new video with a replaced object and a corresponding textual description, ready to be used as a training data point.

B. Temporally-Aware Segmentation: SAM-I2V

We adopt SAM-I2V [9] for promptable video segmentation with temporal awareness and low training overhead. Its design retains a frozen SAM image encoder. It injects temporal coherence via lightweight modules: the Temporal Feature Integrator (TFI), the Memory Selective Associator (MSA), and the Memory Prompt Generator (MPG).

C. First Attempt

Our first attempt (see Fig. 2) utilized an integrated video inpainting pipeline: SAM-I2V provides masks; ControlNet [16] provides structural guidance (depth); IP-Adapter injects reference appearance; AnimateDiff [17] enforces temporal smoothing; and Stable Diffusion [18] inpainting generates the swapped object. This pipeline aimed to modify foreground appearances while preserving underlying behavioral logic.

1) *Critical Failure: Subject–Environment Entanglement:* Despite these efforts, the integrated approach failed in a critical respect: the entanglement between subject and environment. This issue frequently disrupted logical consistency within

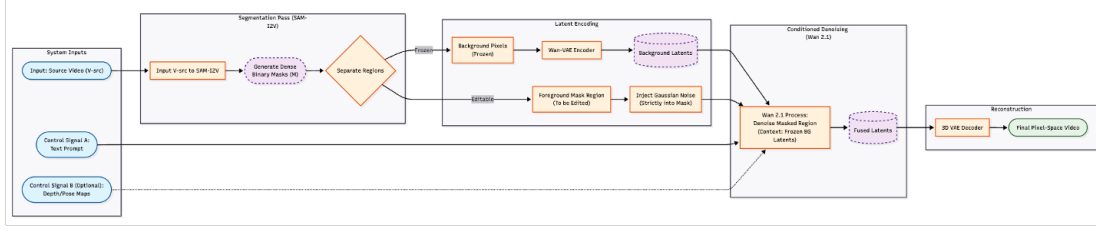


Fig. 3: Object swapping - Second Structure

the augmented videos. We observed two recurring artifacts: **(1) temporal inconsistency** (ghosting/flicker across frames) and **(2) background contamination** (unintended background warping and shifting). These problems became especially severe when the target replacement differed significantly in geometry or scale.

To address entanglement, we redesign the pipeline into two stages see Fig. 3): (1) spatiotemporally stable segmentation, and (2) video-native generation constrained strictly to the masked foreground.

D. Mask-Guided Latent-Space Inpainting

Our final implements (see Fig. 4) designed as a four-step process: **(1) Segmentation pass:** dense masks over all frames; **(2) Latent encoding:** encode video into spatiotemporal latents; **(3) Targeted noise injection:** inject Gaussian noise only into foreground latents (masked region), freeze background latents; **(4) Conditioned denoising & reconstruction:** denoise foreground with semantic prompt (and optional depth/pose), then decode fused latents back to video. This design enforces a hard constraint on background stability and empirically mitigates entanglement artifacts.

E. Video Generation Backbone: Wan 2.1

Wan 2.1 (<https://wan2.video/>) combines a Diffusion Transformer (DiT) with a 3D causal VAE for video-native generation [13]. The 3D VAE enables spatiotemporal compression while preserving temporal continuity, and the DiT backbone effectively captures long-range spatiotemporal relations, supporting scalable, high-fidelity synthesis.

F. VideoQA for Cognitive Feedback

To augment the generated video data with rich, structured descriptions, a Visual Question-Answering pipeline was developed. Using YOLOv8n [19] and Qwen2.5-vl [15] that detects and segments target objects, crops the video to the region of interest, and queries a video-language model to generate structured interaction descriptions.

Its purpose is to generate fine-grained textual descriptions of specific object interactions, providing valuable metadata for training higher-level robotic reasoning. The video is cropped and zoomed to focus exclusively on the segmented target object, removing distracting background elements. The focused video and a text prompt are input into the VLM Model for analysis. The model generates a detailed textual description answering the prompt’s query.

IV. EXPERIMENTS AND RESULTS

A. Object swapping - First Structure

The generated objects often exhibited temporal artifacts, such as translucent remnants from previous frames, which disrupted the illusion of smooth motion. The model struggled to adapt the geometry of a new object (e.g., a small Pikachu) to the motion and volume of the original object (e.g., a large lion), leading to distorted, unrealistic animations. The background would also warp unnaturally in response to the foreground generation.

B. 3D reconstruction

In addition to 2D video generation, the pipeline was extended to 3D reconstruction using the Gaussian-Grouping technique [20]. The goal was to capture a 360-degree video of an object, use SAM-I2V to generate a complete mask, and reconstruct a clean 3D model of the object, isolated from its environment. However, due to significant technical difficulties and project time constraints, this effort was not completed successfully and is categorized as a “failed attempt.”

C. Object swapping - Second Structure

The refined two-stage strategy enables logically consistent transformations, including cross-species replacement and editing clothing attributes, while maintaining a stable background. Across all cases, the generated videos preserve the original gait rhythm and motion trajectory without noticeable drift.

Figure 6 presents four representative examples arranged from top to bottom. In the first row (bear), a walking tiger is transformed into a bear while maintaining coherent body volume and physical presence throughout the sequence. The fur texture adheres to the underlying 3D motion of the animal, and the background remains free of ghosting or warping artifacts, indicating effective spatio-temporal separation between foreground and background.

The second row (cloth) demonstrates editing a clothing attribute on a person. The generated garments respond realistically to scene illumination and cast plausible shadows, avoiding the flat, pasted-on appearance commonly observed in simpler inpainting-based methods.

The third and fourth rows (walking) further evaluate temporal consistency under sustained human motion. In both cases, the model successfully generates new clothing while preserving the original walking dynamics. Although the synthesized fabric exhibits physically plausible secondary motion

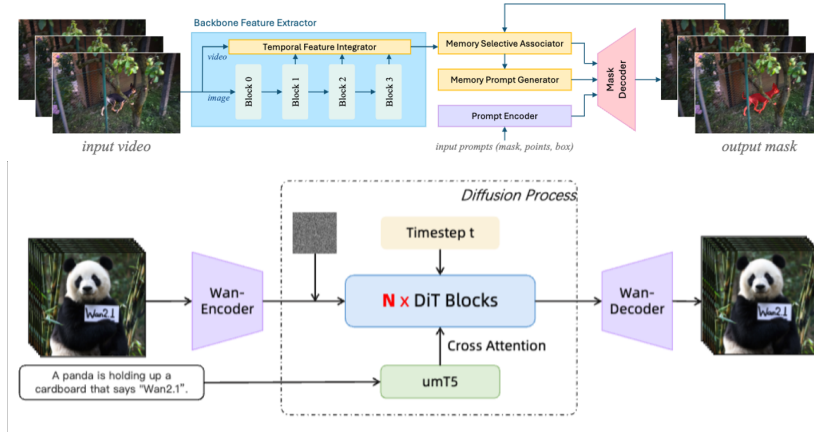


Fig. 4: Core building blocks. Top: SAM-I2V for temporally stable video segmentation. Bottom: Wan 2.1 with a 3D causal VAE and Diffusion Transformer backbone for video generation.

Video Object Analysis Pipeline

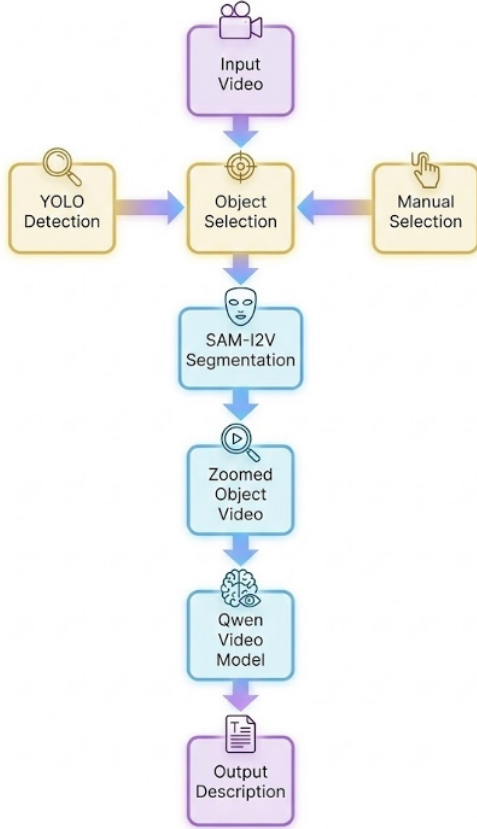


Fig. 5: Object-centric VideoQA pipeline for cognitive feedback and metadata generation.

D. VQA Illustration

We evaluated the effectiveness of the proposed VideoQA pipeline on a basketball video reasoning task (see Fig. 7), where the model must describe the action sequence of an offensive player and the interaction between two selected players. Compared to the baseline approach, which operates directly on the original unprocessed video, our method applies object-centric segmentation and spatial focusing prior to analysis.

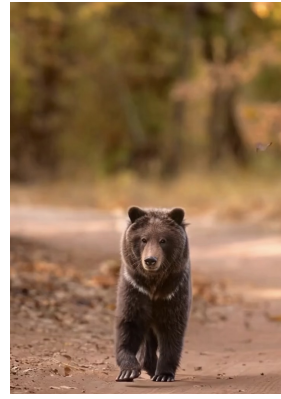
The baseline frequently produces incorrect or incomplete action sequences, failing to capture key events such as dribbling or attempted passes, and produces hallucinated or overly generic descriptions. In contrast, our approach generates accurate, temporally ordered narratives that correctly identify actions such as the initial approach, dribbling, a crossover move, and the final drive toward the basket. The resulting descriptions are also more fine-grained, capturing subtle body dynamics such as weight shifting and the use of the torso to shield the defender. Interaction reasoning is substantially improved: rather than vague statements, our method explicitly describes how the offensive player leverages agility and ball-handling skills, and how the defender attempts to impede progress through positioning and physical contact, including an attempted trip by extending a leg.

Beyond real-world videos, we further apply the VideoQA pipeline to generated clips as a form of qualitative semantic validation. For example, when analyzing a generated animal-swapping result, Qwen2-VL describes the output as a bear walking forward with smooth leg extension, consistent body sway, and a calm facial expression with attentive ear posture. Notably, the model reports no abrupt or unnatural motion, suggesting that the proposed object swapping pipeline preserves kinematic plausibility and temporal continuity without introducing noticeable artifacts.

(e.g., coat sway), the person’s underlying gait, rhythm, and trajectory remain locked to the source video, demonstrating strong preservation of kinematic consistency.



(a) Original



(b) Generated



(c) Original



(d) Generated



(e) Original



(f) Generated



(g) Original



(h) Generated

Fig. 6: Qualitative results (four cases). Each row shows the original input (left) and the generated output (right).

Baseline Response

In this sequence of frames, the offensive player, wearing a dark jersey, is dribbling the ball while being closely guarded by a defensive player in a white jersey. Here's a breakdown of the moves and interactions:

1. **Initial Dribble:** The offensive player begins with a quick dribble, trying to create space between himself and the defender.
2. **Passing Option:** As he dribbles, he looks for an opening or a teammate who might be open for a pass.
3. **Defensive Pressure:** The defender in the white jersey is actively trying to block the offensive player's path and disrupt his movement.
4. **Transition to a Drive:** The offensive player then transitions into a drive towards the basket, attempting to get past the defender.
5. **Defensive Reaction:** The defender tries to block the offensive player's progress, but the offensive player continues to maneuver around him.

The interaction between the offensive and defensive players is intense, with the offensive player using his agility and ball-handling skills to navigate through the defense, while the defender uses his positioning and physicality to try and stop him.

Ours (Proposed Method)

The offensive player, dressed in a dark outfit with a green headband, executes a series of moves to maneuver around the defender. Here's a breakdown of the sequence:

1. **Initial Approach:** The offensive player starts by moving forward, attempting to create space and position themselves for an attack.
2. **Dribbling:** The player dribbles the ball while maintaining control, shifting their weight and body to evade the defender.
3. **Crossover Move:** The offensive player performs a crossover move, where they switch the ball from one hand to the other while stepping over the defender's outstretched leg.
4. **Drive to the Basket:** After the crossover, the player drives towards the basket, using their momentum to attempt a shot or pass.

Interaction Between Players:

- The defender, wearing a white outfit, tries to block the offensive player's path and disrupt their movement.
- As the offensive player executes the crossover, the defender attempts to react by moving their legs and arms to try and regain control of the play.
- The offensive player's quick footwork and ball-handling skills help them maintain possession and progress towards the basket despite the defender's efforts to defend.

Fig. 7: VideoQA analysis yields more accurate and better-ordered interaction descriptions than raw-video prompting.

V. CONCLUSION AND FUTURE WORK

This project presents a practical pipeline for generative semantic augmentation in robot vision, addressing the challenges of data scarcity and domain shift in video-based perception. These can be summarized as follows:

- We successfully integrated SAM-I2V to achieve precise, temporally stable video object segmentation, providing a reliable foundation for isolating foreground objects from their backgrounds.
- We implemented a generative semantic augmentation pipeline via object swapping. Through an iterative design process, the final architecture combining SAM-I2V and Wan 2.1 effectively resolves subject–environment entanglement and logical consistency issues that commonly hinder traditional video inpainting approaches.
- We developed a supplementary VideoQA system leveraging YOLO and the Qwen video model, capable of producing accurate, fine-grained cognitive reasoning and structured textual descriptions for both original and generated video data.

Building on the success of the current pipeline, we identify three primary directions for future research aimed at further automating data generation and bridging the gap between simulation and real-world robotic applications:

- **VQA as a Validation Loop:** We plan to integrate the VideoQA module as an automated validation mechanism. By analyzing generated textual descriptions programmatically, the system can filter out low-quality or physically

implausible videos, improving dataset reliability and mitigating model hallucinations.

- **Sim-to-Real Transfer:** A key future objective is to deploy the proposed pipeline for generating training data for physical robots. Empirical evaluations will assess whether models trained with generative semantic augmentation demonstrate improved performance and generalization in real-world manipulation and navigation tasks.
- **Vision–Language–Action Integration:** In the long term, we aim to extend the VideoQA module beyond descriptive analysis toward action grounding. By translating textual reasoning into executable robot commands, the system could enable a closed-loop perception-to-action framework, advancing the development of more autonomous and intelligent robotic systems.

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